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Original Research

Post-match recovery of eccentric knee flexor strength in male professional football players

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ABSTRACT

Objectives: This study aimed at verifying the effect of a football match on the eccentric knee flexor strength of male professional players along a 72-h period.
Study design: Quasi-experimental study.
Setting: Football club facilities.
Participants: Fifteen players were assessed in four timepoints: 24 h before, immediately after, 48 h and 72 h after the match.
Main outcome measures: The eccentric knee flexor strength was assessed during the Nordic hamstring exercise execution.
Results: Players presented a significant strength reduction immediately after match ($\Delta = 12\%$; $p = 0.001$; large effect size, $d = 1.10$), and did not recover their strength capacity within a 48 h-period ($\Delta = 6\%$; $p = 0.011$; moderate effect size, $d = 0.57$). At 72 h after the match, strength was similar to baseline levels ($\Delta = 3.5\%$; $p = 0.122$; small effect size, $d = 0.34$). Secondarily, individual response analysis considered a player 'fully recovered' when his strength deficit compared to the baseline measure was lower than the measurement coefficient of variation (i.e., <5%). Only 6 (40%) and 9 (60%) players were 'fully recovered' at 48 h and 72 h after the match, respectively.
Conclusion: Professional football players experienced an immediately post-match drop on the eccentric knee flexor strength, and significant strength deficits persisted for a 48-h period. Some players were not recovered at 72 h after the match.

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1. Introduction

Hamstring strain injury (HSI) is the most prevalent non-contact injury in men's professional football (soccer) (Ekstrand et al., 2020). The recurrence rate is high (13% within 2 months of return to play) (Ekstrand et al., 2016), and the average time lost due to re-injuries is even longer than index injuries (21.5 *vs.*18 days) (Ekstrand et al., 2020). Consequently, the burden of HSI (i.e., the cross-product of severity and incidence) harms the clubs' performance and finances (Hägglund et al., 2013; Bahr et al., 2018). Preventing HSI is a primary concern of medical/coaching staffs (Bahr et al., 2018), and elite football clubs continually invest in screening tests and preventive strategies to minimize this issue (McCall et al., 2014; Meurer et al., 2017). Some prospective cohort studies with football players (Croisier et al., 2008; Fousekis et al., 2011; Lee et al., 2017; Timmins et al., 2016), but not all (van Dyk et al., 2017), have supported the low eccentric knee flexor strength as a risk factor for HSI. Therefore, eccentric knee flexor strength has been largely assessed in football players using isokinetic dynamometers (Croisier et al., 2008; Fousekis et al., 2011; Lee et al., 2017; Ribeiro-Alvares et al., 2019; van Dyk et al., 2017) or load cells systems during the Nordic hamstring exercise (NHE) execution (Ribeiro-Alvares et al., 2020; Timmins et al., 2016; van Dyk et al., 2017).

Throughout a 90-min football match, male professional players cover an average total distance close to 11 km of which 350 m and 1150 m are performed in sprints and high-intensity running,

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respectively (Barnes et al., 2014). Match demands include dozens of specific actions, such as accelerations, decelerations, cuttings, jumps, passes, and shots (Bradley et al., 2013). Consequently, muscle fatigue, characterized as an involuntary exercise-induced decline of physical performance (including temporary reduction of muscle strength), is an expected response to football matches (Silva et al., 2018). Studies on intermittent exercise protocols designed to mimic demands of competitive football matches have found significant decreases on the eccentric knee flexor strength of amateur (Jones et al., 2015), semi-professional (Lovell et al., 2013), and professional players (Rhodes et al., 2019). More hamstring strain injuries appear to occur in the final third of both match halves (Woods et al., 2004), suggesting that fatigue might play a role in the multifactorial causation of this injury in football, which is in line with perceptions of elite team practitioners (McCall et al., 2014; Meurer et al., 2017).

Although the International Olympic Committee consensus suggests that football matches should be interspersed by at least 96 h to protect players from injury (Soligard et al., 2016), it is not uncommon players being exposed to two matches within a 72-h interval. This may lead to 'residual fatigue', a phenomenon characterized by physiological stress and possible impairments in the players' physical performance (Silva et al., 2018). Fixture congestion periods are also associated with an increase in muscle injury rates (Bengtsson et al., 2018), and a reasonable hypothesis is related to the insufficient muscular recovery between matches. However, few studies have assessed the post-match recovery of knee flexor strength in football players (Rhodes et al., 2019; Silva et al., 2013; Wollin et al., 2017, 2018). Some Some studies have reported that knee flexors isometric and concentric strengths of football teams were fully restored within 48 h after a match (Silva et al., 2013; Wollin et al., 2017, 2018). Conversely, eccentric knee flexor strength deficits seem to persist up to 72 h after a simulated football match (Rhodes et al., 2019).

Considering the high eccentric demand on the hamstring muscles during a football match (Akenhead et al., 2013), as well as the association of knee flexor eccentric strength with HSI (Croisier et al., 2008; Fousekis et al., 2011; Lee et al., 2017; Timmins et al., 2016), the recovery of knee flexor eccentric strength after football matches needs to be investigated. To the best of our knowledge, there is a lack of investigations on the recovery of eccentric knee flexor strength of professional football players after a real match. Moreover, previous studies have focused only on the average recovery status of the team, neglecting the individuality of biological responses. Therefore, the present study aimed to verify the effect of a football match on the eccentric knee flexor strength of male professional players along a 72-h period. Secondarily, we aimed to verify the individual responsiveness of players' strength recovery.

2. Methods

2.1. Study design

This quasi-experimental study included professional male players of a football club competing at first state championship level in Brazil. In the last week of preseason period, this club performed two friendly matches in a single day (Saturday) to expose most of the squad to a match demand before the first official match of the season. Each participant was assigned to only one match. No training session was performed on Sunday, and a single regenerative training session (i.e., slow-intensity running and mobility exercises) was performed on Monday. Participants had their eccentric knee flexor strength assessed in four timepoints: 24 h before match (baseline assessment; Friday), immediately after match (Saturday), 48 h (Monday) and 72 h (Tuesday) after match. Both matches and evaluations took place in the club facilities. The study was approved by the institutional ethics committee (protocol number: 2.465.908), and all participants provided written informed consent.

2.2. Participants

Twenty professional male football players were initially engaged in this study (i.e., all players scheduled to start the two friendly matches, except for goalkeepers). All of them were over 18 years old, had current professional contracts with the club, and were actively participating in the team's training routine. Participants with injuries were automatically excluded from friendly matches, while those returning from periods of absence due to injury or any other reason played only in the second half and were not included in this study. Additionally, only participants who played at least 80 min were included in the statistical analysis.

Matches analyzed in this study were performed at the 5th week of preseason. The first week (only three work days) was used to carry out the routine preseason tests. Along the four subsequent weeks, participants trained from Mondays to Saturdays, including 1–2 strength training sessions per week with the NHE (1 set of 8 repetitions per session). After the 2nd training week, the team played the first preseason friendly match. After the 4th training week, the two friendly matches assessed in the present study were played. Along the 4th training week, participants performed a strength/power training session (including the NHE) on Monday (i.e., four days before baseline evaluation), and technical/tactical practices on the other days of the week. Thursday and Friday training sessions were designed with low volume and intensity, aiming to preserve participants for the Saturday's matches. Baseline evaluations were carried out immediately before Friday's training session.

2.3. Eccentric knee flexor strength test

Evaluations were preceded by a general warm-up structured and guided by the team strength and conditioning coach (except in the post-match evaluation). All evaluations were carried out by the same pair of investigators, both with a large experience with the procedures. The eccentric knee flexor strength was assessed during the NHE execution on a custom-made device built by our research group (Fig. 1) and used in previous studies (Oliveira et al., 2020; Ribeiro-Alvares et al., 2020). This device was based on the prototype validated by Opar et al. (2013), and increasingly used around the world (Buchheit et al., 2016; van Dyk et al., 2017). As the NHE was included in the team strength training program, all participants were already familiar with this exercise.

Participants started the test in kneeling position over a padded board (with the hip neutral and the torso upright). They were encouraged to slowly lower their torso to the floor level in a constant cadence (~3 s per repetition) using only the knee joint (i.e., maintaining the hips and spine neutral), with their arms flexed at the elbow joints facing the palms of the hands forward at the level of the shoulders. Participants were instructed to use their upper limbs to absorb the fall and to return to the initial position.

A single researcher supervised the correct execution of the NHE, controlling the execution time and verbally encouraging the participants to continue eccentrically contracting their hamstrings as long as possible. Participants performed 3 valid repetitions of the NHE, with a minimum 10-s interval between them. A repetition was considered valid when the participant performed a slow and steady movement in ~3 s or more, with no apparent flexion/extension movements of hip or lumbar spine, and no apparent side compensation. No more than five trials were needed for each participant.

Fig. 1. Assessment of eccentric knee flexor strength during the Nordic hamstring exercise execution.

During the NHE execution, the left and right ankles were individually secured by ankle braces placed superior to the lateral malleolus and fixed to two independent commercially-available load cells (E-lastic, *E-sporte Soluções Esportivas*, Brazil; sample rate = 10 Hz) perpendicularly attached to the board. Both load cells were calibrated before each data collection session. Force data were simultaneously transferred via Bluetooth to two smartphones. Force values obtained in each limb were registered in every repetition. The peak values were used to calculate the between-limb average value used for data analysis. Test-retest reliability with 95% confidence intervals of our device was assessed through data from 20 healthy active men obtained in our laboratory: test value, 298.5 N (257.7–321.2 N); retest value, 297.7 N (265.9-329-4 N); intraclass correlation coefficient, 0.96 (0.91–0.99); typical error, 14.8 N (11.2–21.9 N); coefficient of variation, 4.7% (3.5–7.0%); minimal detectable change (MDC), 41 N and 14.2%.

2.4. Statistical analysis

After distribution normality have been confirmed by Shapiro-Wilk test, a repeated measures ANOVA was conducted to compare mean differences between timepoints. Mauchly's test was used to check data sphericity, and the Greenhouse-Geisser correction was applied. A Fisher's Least Significant Difference (LSD) post-hoc analysis was conducted to determine which timepoints were different. The level of significance was set at $p < 0.05$. Effect sizes (ES) between baseline and post-match measures were calculated using Cohen's d, and ES were considered as "trivial" (ES < 0.2), "small" (ES > 0.2), "medium" (ES > 0.5) or "large" (ES > 0.8) (Cohen, 1988). A recovery classification was created by authors to assess the individual responsiveness based on strength deficit at 48 h and 72 h in relation to baseline measure: 'fully recovered', when strength deficit was smaller than coefficient of variation (CV; i.e., deficit <5%); 'partially recovered', when strength deficit was higher than CV (i.e., deficit >5%); and 'not recovered', when strength deficit was higher than minimal detectable change (MDC; i.e., deficit >14%).

3. Results

Twenty players were initially engaged in this study. Five participants were replaced during matches. These participants performed <80 min and were excluded from the study. Therefore, 15 participants (age, 25.73 ± 4.14 years; height, 1.81 ± 06.61 m; body mass, 76.47 ± 8.14 kg; percent body fat, 15.41 ± 2.85%) completed the schedule. No participant had a history of HSI in the prior season (i.e., at least 12 months free of HSI), and two participants had older HSI in the right limb (without apparent sequelae on strength production). Fourteen participants played the full match (i.e., ~90 min), while one participant was replaced in the 85th minute. There was no injury or reports of pain/discomfort while performing the NHE tests.

There was a significant time effect ($F = 9.379$; $p = 0.001$). th A significant reduction of eccentric knee flexor stregnth was observed immediately after match ($\Delta = 12\%$; $p = 0.001$; large ES, $d = 1.10$). Baseline strength capacity was not recovered up to 48 h after match ($\Delta = 6\%$; $p = 0.011$; moderate ES, $d = 0.57$). At 72 h after match, the eccentric knee flexor strength was similar to baseline levels ($\Delta = 3.5\%$; $p = 0.122$; small ES, $d = 0.34$). These results are expressed in Fig. 2.

Individual response analysis showed that only 6 (40%) and 9 (60%) participans were 'fully recovered' (strength deficit <5%) at 48 h and 72 h after match, respectively. Two participants (13%) were classified as 'not recovered' up to 72 h after match. The individual responses are illustrated in Fig. 3.

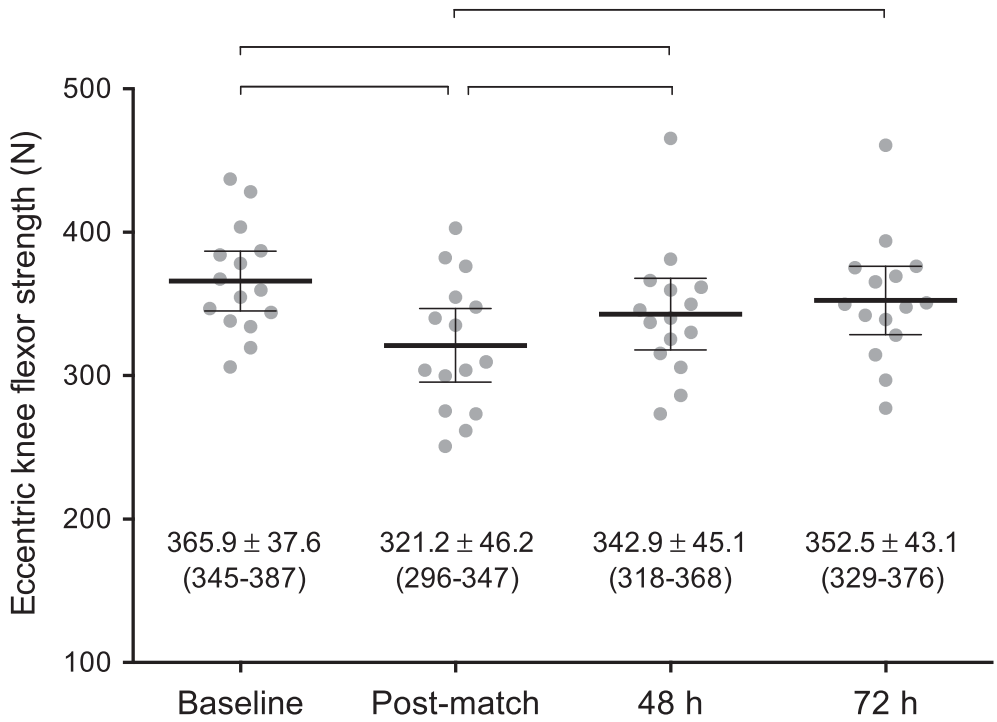

Fig. 2. Eccentric knee flexor strength absolute values [Mean ± SD (95% CI)]. Gray circles represent individual performances. Brackets denote significant difference between timepoints ($p < 0.05$).

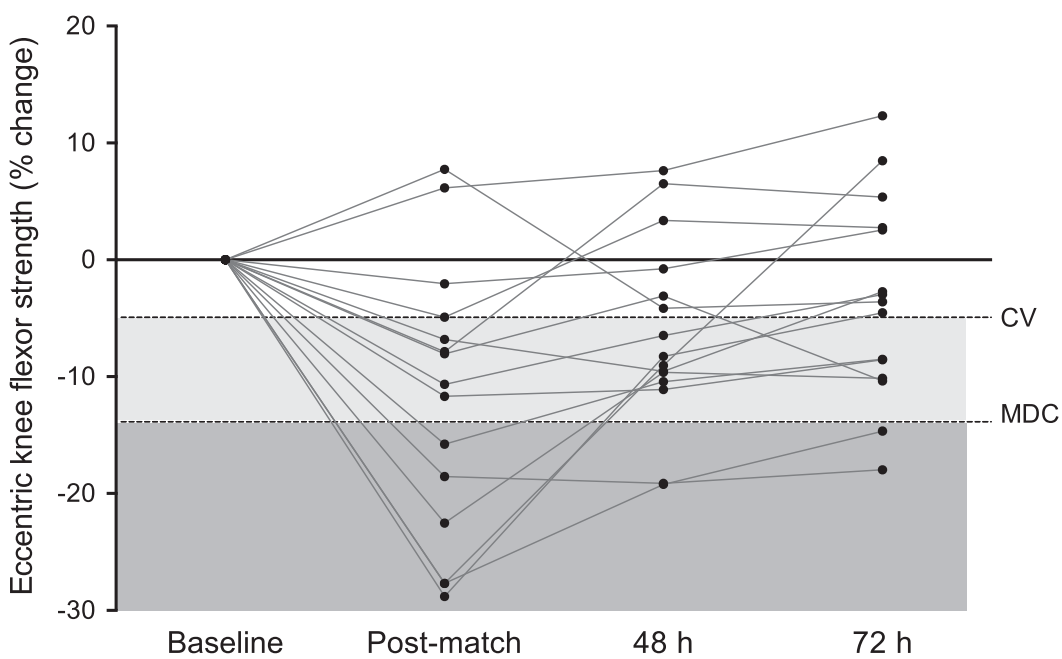

Fig. 3. Players' relative responses of eccentric knee flexor strength (data normalized by the baseline measure). Light gray zone denotes the 'partially recoverd' players (strength deficit higher than coefficient of variation - CV), while dark gray zone denotes the 'not recovered' players (strength deficit higher than minimal detectable change - MDC).

4. Discussion

To our best knowledge, this is the first study assessing the time course of eccentric knee flexor strength recovery after a football match in professional players. The main findings of this study were that the group of players experienced persistent knee flexor eccentric strength deficits up to 48 h after match. At 72 h after match, the group had strength levels similar to baseline; however, 6 players (i.e., 40%) were not considered 'fully recovered' within this time frame.

Silva et al. (2013) reported no changes on the isokinetic concentric knee flexor strength after a professional-level football match. Wollin et al. (2017) demonstrated that youth (~17 years old) football players presented a significant isometric strength deficit

(~5%) at 24 h after match, and restored their knee flexor strength capacity at 48 h. Rhodes et al. (2019) reported significant isokinetic eccentric strength deficits in professional football players up to 72 h after a simulated football match protocol. Adding with the findings of the current study, eccentric knee flexor strength seems to have a more time-consuming recovery pattern compared to concentric (Silva et al., 2013) and isometric (Wollin et al., 2017) strength after a football match. However, given the absence of studies assessing post-match recovery in different types of contraction, this hypothesis still needs to be tested. Other confounding factors, such as age, competitive level, physical conditioning, period of the season, and tools used to assess knee flexor strength (including its typical error of measurements), may be related with the inconsistent results reported so far.

Despite inconsistent findings in the literature (Green et al., 2020), eccentric knee flexor strength has been associated with a high HSI risk by prospective cohort studies involving football players (Croisier et al., 2008; Fousekis et al., 2011; Lee et al., 2017; Timmins et al., 2016) and other athletes (Opar et al., 2015; Sugiura et al., 2008). According to the multivariate logistic regression model by Timmins et al. (2016), the risk of a professional football player sustaining an HSI increases ~9% at every 10 N decrease in eccentric knee flexor strength. Considering that players in the current study presented an average drop of ~23 N and ~13 N at 48 h and 72 h after match, it is reasonable to assume that players would be more susceptible to HSI along this period. Notwithstanding, players are often exposed to two matches within a 3-day interval in some countries. In Brazil, where the current study was conducted, elite teams involved in continental cups sometimes plays 70–80 matches per year, including several weeks with two matches in a period shorter than 72-h (e.g., Wednesday night match followed by Saturday afternoon match). A 14-year prospective cohort study evidenced that muscle injury rates in professional football were significantly lower in matches preceded by 6 or more days compared with 3 or fewer days since the previous match exposure (Bengtsson et al., 2018). Therefore, the persistent strength deficits presented by some players in the current study (see Fig. 3) may be speculated as a possible contributing factor to the higher incidence of HSI when matches are played within a 3-day interval.

The risk of sustaining an HSI is nine-fold greater during matches than in training sessions (Ekstrand et al., 2016). However, even without a congested week, players are at risk of sustaining an HSI due to the training exposure. Exposing players to large and rapid increases in high-speed running distances increased the odds of injury (Malone et al., 2018). Conversely, players who complete moderate high-speed running distances seem to be at reduced injury risk compared to those with lower training workload (Malone et al., 2018). Therefore, the management of players' workload along the season has been accepted as a key element for preventing HSI in professional football (Buckthorpe et al., 2018). A recent cluster-randomized controlled trial with youth footballers found no preventive effect of workload management using the acute:chronic workload ratio (Dalen-Lorentsen et al., 2020), highlighting the need of more research on this topic. Our findings add that post-match recovery is not homogeneous among players. In theory, those players with poor post-match recovery would be more susceptible to sustaining an HSI because the training workload would be applied to a muscle still in the recovery process. Therefore, the management of week training routine should consider not only factors as the acute and chronic workloads but also the post-match recovery pattern presented by each player.

In the football context, post-match recovery is usually assessed through subjective ratings (Nedelec et al., 2014) or biochemical analysis (Silva et al., 2013). Up to now, there is no evidence to support the use of these measures as an injury prevention strategy. Conversely, monitoring the players' strength status after matches emerges as a promising perspective for preventing HSI. This is not an easy task due to the short time to assess a group of players within a schedule of professional football that includes matches, training sessions, travels, etc. However, Wollin et al. (2020) monitored the isometric knee flexor strength on the first regular training day back after a rest day (≥40 h post-match), and individual training workload management was applied to players with strength deficits. As a result, authors found a reduction of ~80% in the HSI rate when compared with the previous seasons without strength monitoring (Wollin et al., 2020). Considering the association of eccentric knee flexor strength with HSI (Croisier et al., 2008; Fousekis et al., 2011; Lee et al., 2017; Timmins et al., 2016), it seems reasonable that this secondary prevention strategy can positively impact the HSI rate of a team using the strength testing adopted in our study. However, we should consider that the maximum eccentric test with the NHE can be more stressful for athletes than the isometric test, requiring further attention in its insertion into the team's training routine.

Some limitations of this study should be acknowledged. First, information regarding the match demands (i.e., internal and external workload measures) and post-match issues (e.g., sleep time, hydration status, diet, life habits) would allow a deeper investigation on the individual responses. Second, since we had only one testing device, some players performed the strength assessment immediately after match with a longer rest time than others (<10 min between the first and the last participant). Even so, the discussion was focused on the follow-up period (48 and 72 h after match) due to its potential impact on HSI risk in exposures (training or matches) performed within the three subsequent days. Third, note this study was conducted surrounding friendly matches at late preseason. Post-match recovery may be improved throughout the season in response to a conditioning process, as well as may be impaired in congested periods and/or after decisive matches with higher physical demands. Lastly, despite the number of players assessed in this study have been similar (Mohr et al., 2016; Rhodes et al., 2019; Wollin et al., 2017, 2018) or greater (Lovell et al., 2013; Nedelec et al., 2014) than the existing literature, it must be assumed that this is not a large-sample study and results should be treated with caution in terms of external validity.

As strengths, this study was carried out within the 'real-world' of professional football. The testing device is cheap and portable. The data collection procedures are quick, requiring no more than two staff members, and can be carried out inside the club's facilities (e.g., in the locker room). This means the method proposed in this study is feasible to be implemented by football clubs interested in monitoring their players' eccentric knee flexor strength.

5. Conclusion

Professional football players experienced an immediately post-match drop on the eccentric knee flexor strength, and significant strength deficits persisted for a 48-h period. Some players were not fully recovered at 72 h after match. Those players with poor post-match recovery would be theoretically exposed to a higher risk of sustaining an HSI in the days following the match. Therefore, monitoring the individual strength recovery status seems advisable when designing the subsequent training sessions, as well as to expose the player to a next match within a 3-day period.

Ethical statement

The study was approved by the Ethics in Research Committee of the Federal University of Health Sciences of Porto Alegre, Porto Alegre, RS, Brazil (approval number: 2.465.908).

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Declaration of competing interest

Authors declared no conflict of interest.

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